LIÈGE université
Sciences Appliquées

ELEN0449

NARUTO HAND SIGN RECOGNITION

REAL-TIME GESTURE RECOGNITION PIPELINE

PRESENTED BY:

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MOTIVATION

THE TWELVE CLASSES

A ZODIAC OF FINGER-CONFIGURATIONS.

Each sign is named after a Chinese-zodiac animal.

PROBLEM STATEMENT

CLASSIFY WHICH OF **THE TWELVE MUDRA** A PLAYER PERFORMS – FROM A LIVE WEBCAM.

INPUT

Live RGB video

1280 × 720 · 30 fps

Standard built-in webcam. No depth sensor, no glove, no marker.

OUTPUT

$\hat{y} \in \{1 \ldots 12\}$

1per-frame prediction

CONSTRAINT

REAL-TIME

Latency budget set by 30 fps gameplay

DATASET – COLLECTION PROTOCOL

NO PUBLIC SET FIT THE TASK – SO **WE RECORDED OUR OWN.**

PROTOCOL

1. Show participant a reference image of the target mudra.
2. Record ~10 s at 30 fps facing the webcam.
3. Encourage gentle sideways drift for viewpoint variability.
4. Discard the first 2 s (transition / settling noise).
5. Repeat for all 12 classes.

TOTAL RAW FRAMES

39,600

11 participants × 12 classes × 300 frames

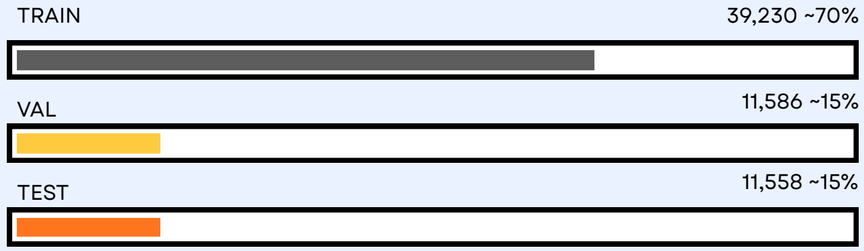
PROTOCOL

11 *- PEOPLE*

7♂.4♀ *- GENDER SPLIT*

VARIED *- LOCATIONS / LIGHT*

PARTICIPANT-WISE SPLIT

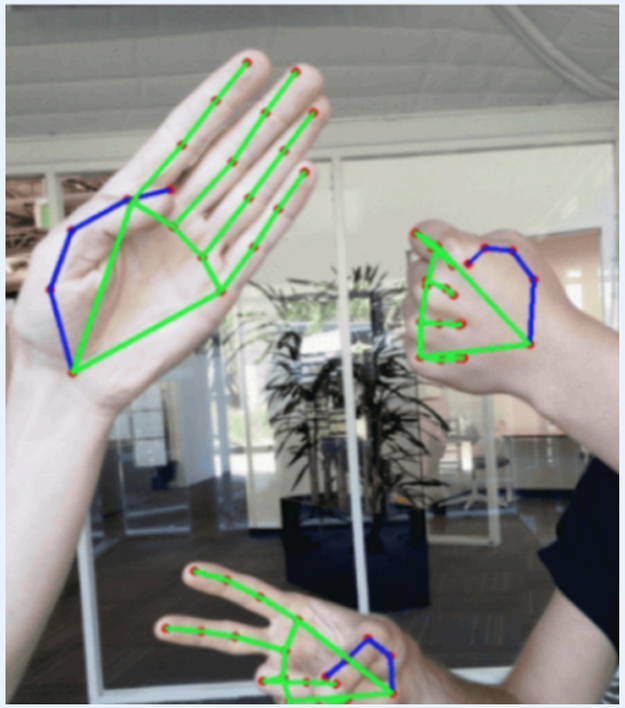


// all frames from a given participant stay in one split – no leakage

FROM PIXELS TO LANDMARKS – MEDIAPIPE

A MILLION PIXELS COLLAPSE INTO **126 NUMBERS.**

21 KEYPOINTS / HAND



// 21 landmarks · 3D world coordinates · per hand

PER-FRAME FEATURE VECTOR

2 x 21 x 3 = 126

HANDS LANDMARKS XYZ FEATURES

WHY LANDMARKS, NOT PIXELS

- Robust to lighting, skin colour, background.
- Robust to translation & camera distance.
- Preserves the geometry that actually discriminates signs.
- Tiny inputs – works on a modest dataset.

// CAVEAT

When two hands touch (Rat, Snake, Tiger…) MediaPipe occasionally drops one hand. We don't repair missing landmarks – the classifier learns the gap.

PIPELINE OVERVIEW

-INPUT

RAW FRAME

H x W x 3

webcam . 30fps

-

MEDIAPIPE

126 features

MudraMLP

126 > 256 > 128 >12

DecisionTree

-OUTPUT

$\hat{y} \in \{1...12\}$

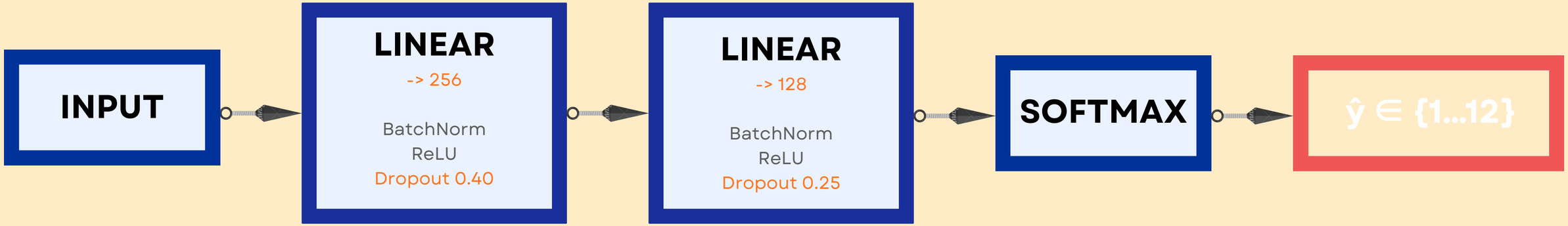
per-frame label

// DataSet Augmentation

we added a preprocessing step, where we apply mirroring to make our dataset more bigger

MODEL ARCHITECTURE

MUDRAMLP – SMALL, DENSE, REGULARISED.



2
– HIDDEN LAYERS

~50k
– Parameters

12
– Output classes

TRAINING SETUP

OPTIMISATION

Optimiser	Adam
Learning rate	5×10^{-4}
Weight decay	1×10^{-4}
Batch	size128
Loss	class-weighted CE

REGULARISATION

ReduceLROnPlateau
factor ½ · after 5 stalled val epochs

Early stopping
patience = 15 epochs

Checkpoint
best by val-loss → kept as final model

Augmentation
mirror flip → +13.9% test acc.

// MIRROR AUGMENTATION

Doubling training diversity by flipping hand configurations & swapping left/right roles produced more physically valid sign variants at zero recording cost.

-Decision Tree
67.5% → **83.7%**

- MUDRA-MLP
81.9% → **93.7%**

RESULTS

Per class results

LABELS	COUNT	TP		TN		FN		FP		TOTAL
	#	#	%	#	%	#	%	#	%	#
bird	966	964	99.8%	10585	91.6%	2	0.02%	7	0.72%	11558
boar	966	966	100.0%	10569	91.4%	0	0.00%	23	2.33%	11558
dog	966	895	92.7%	10581	91.5%	71	0.61%	11	1.21%	11558
dragon	966	966	100.0%	10580	91.5%	0	0.00%	12	1.23%	11558
hare	964	960	99.6%	10594	91.7%	4	0.03%	0	0.00%	11558
horse	964	964	100.0%	10594	91.7%	0	0.00%	0	0.00%	11558
monkey	954	924	96.9%	10560	91.4%	30	0.26%	44	4.55%	11558
ox	946	902	95.3%	10566	91.4%	44	0.38%	46	4.85%	11558
ram	964	911	94.5%	10548	91.3%	53	0.46%	46	4.81%	11558
rat	974	974	100.0%	10344	89.5%	0	0.00%	240	19.77%	11558
snake	960	718	74.8%	10593	91.7%	242	2.09%	5	0.69%	11558
tiger	968	924	95.5%	10534	91.1%	44	0.38%	56	5.71%	11558

Headline

+90% Accuracy

Perfect Class

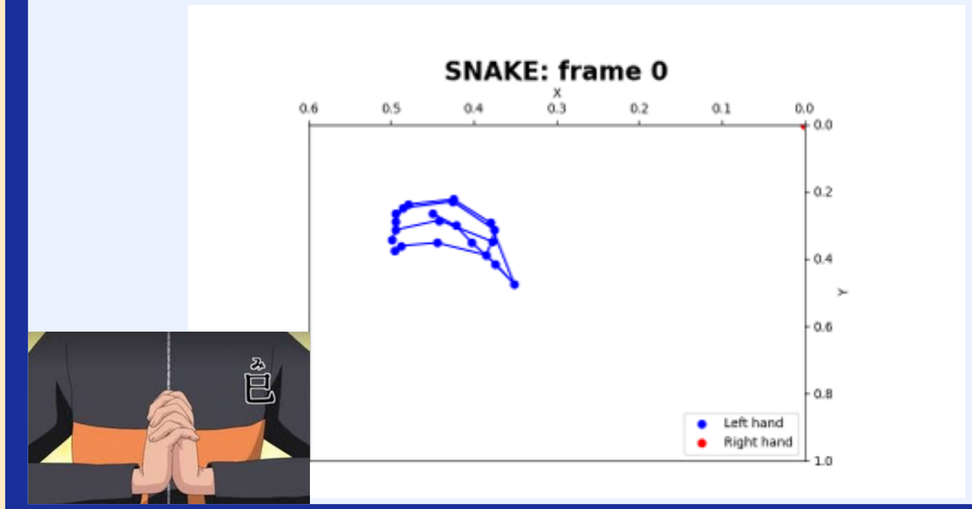
100% Horse

// Failure

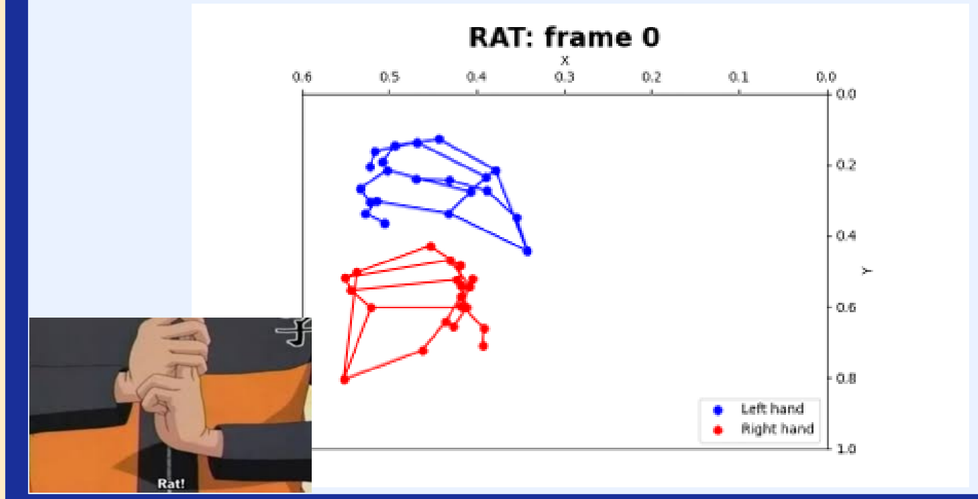
Snake confused with rat 25% of the time

FAILURE ANALYSIS

Detected - Snake



Detected - rat



// DIAGNOSIS

When MediaPipe drops a hand, the Snake feature vector collapses into something visually indistinguishable from half a Rat – the classifier inherits the upstream ambiguity.

DEMO

BE READY TO HAVE SOME FUN!

CONCLUSION

WHAT WORKS

WHAT’s NEXT

THANK YOU